

Roll Number: BT2024185

GitHub: https://github.com/Sairangareddy2006/ML_Assignment

Assignment 1: Polynomial Regression Report

Geothermal Power Plant Expansion & Reservoir Mapping

1. Introduction

This report presents polynomial regression models developed for a multi-stage geothermal power plant expansion project using roll number BT2024185 datasets:

- Phase 1: Steam Turbine Optimization (var1): Predict Net Power Score (y) from 6 turbine operating parameters (x1 to x6, deviations bounded in [-1, 1]). Candidate models span up to degree 10.
- Phase 2: Subterranean Thermal Reservoir Mapping (var2): Predict Thermal Anomaly Score (y) across 3D survey coordinates (x1, x2, x3 in [-1, 1]). Candidate models span up to degree 20.

Models are evaluated using 5-fold cross-validation with Mean Squared Error (MSE) and Coefficient of Determination (R2). Polynomial features are normalized with StandardScaler to avoid numerical scale discrepancies.

2. Phase 1: Steam Turbine Optimization (var1)

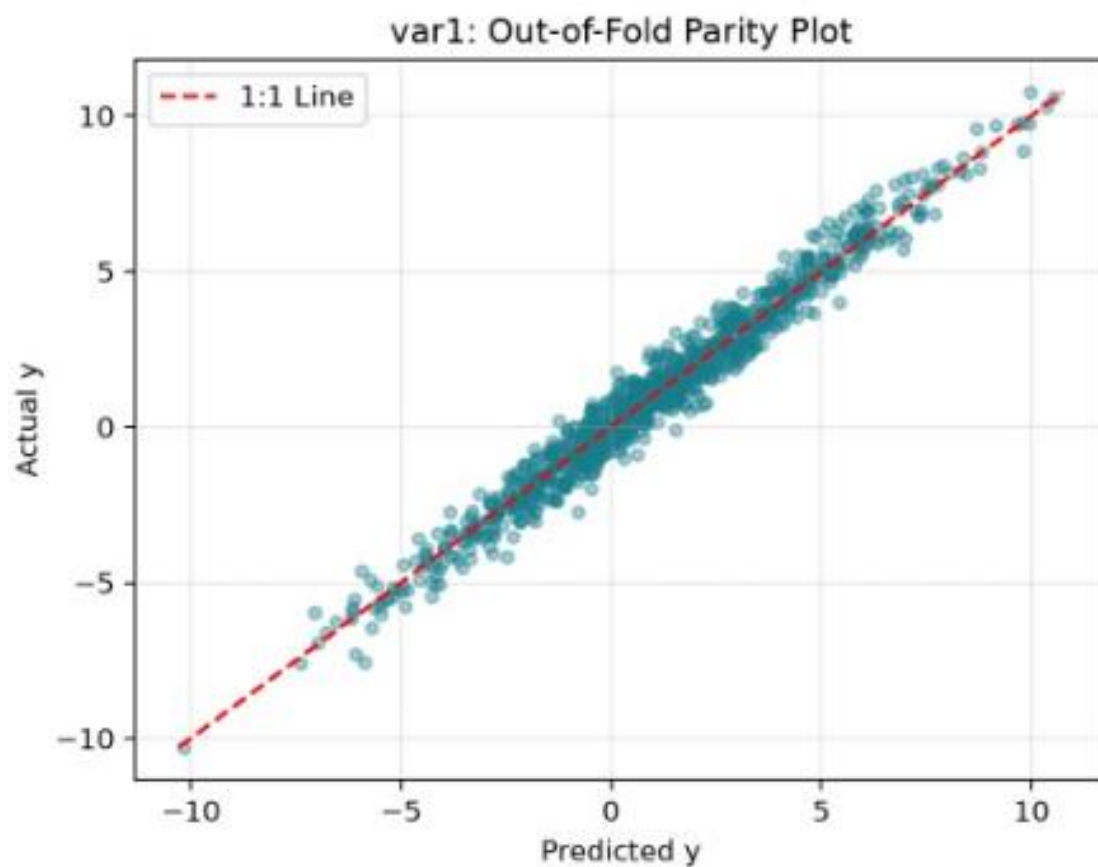
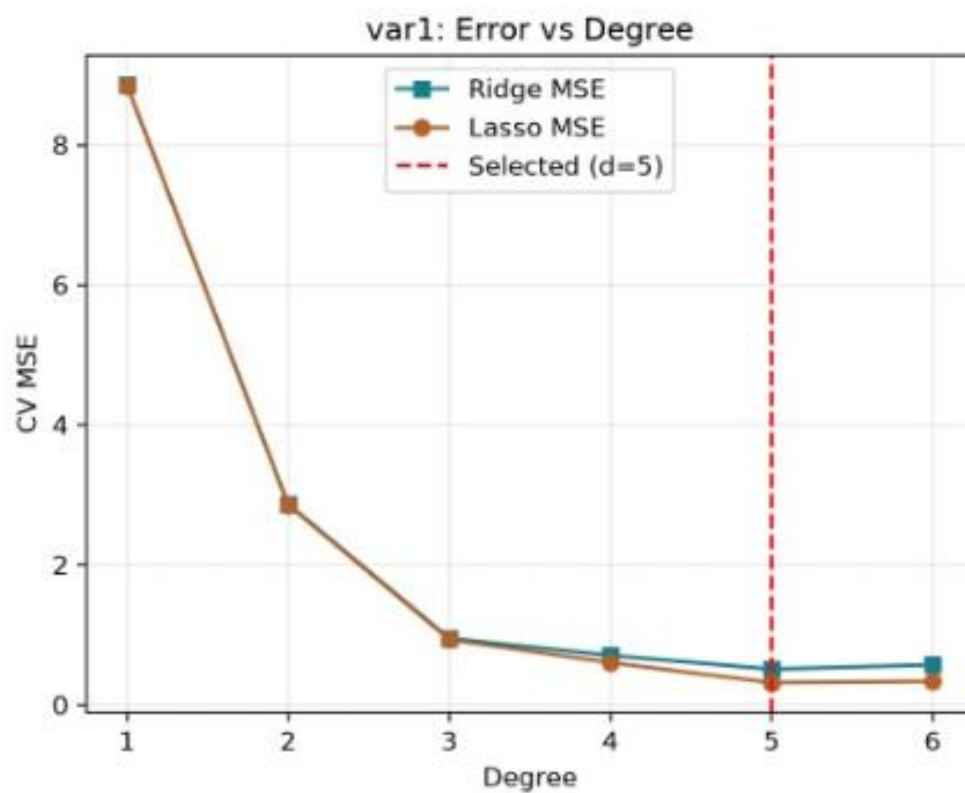
Data Inspection and Linear Baseline

The dataset contains 1,000 train and 1,000 test observations with no missing values. Target y ranges from -10.28 to +10.76 (mean 1.07, std 3.14). Individual input correlations with y are weak ($|r| \leq 0.23$), indicating that single variables cannot explain power output without non-linear interactions. A baseline linear regression model (Degree 1) achieved CV MSE = 8.8554 and R2 = 0.0954 (explaining just 9.5% of variance). This high bias confirms that a linear hyperplane severely underfits turbine aerodynamics.

Degree Exploration and Model Selection

I evaluated polynomial expansions from degree 1 to 6 using Ordinary Least Squares (OLS), Ridge (L2), and Lasso (L1) with 5-fold cross-validation.

Degree	Features	Ridge Val MSE	Ridge Val R2	Lasso Val MSE	Lasso Val R2
1	6	8.855	0.095	8.846	0.096
2	27	2.869	0.704	2.855	0.706
3	83	0.950	0.901	0.932	0.902
4	209	0.705	0.927	0.603	0.937
5	461	0.509	0.947	0.3154	0.9672
6	923	0.570	0.941	0.3341	0.9652



As shown in Table 1, unregularized OLS begins overfitting past degree 4, and error grows as features approach sample size. Lasso (L1) resolved this by zeroing out 347 redundant interaction terms, retaining 114 active features. This achieved the minimum CV MSE of 0.3168 ($R^2 = 96.78\%$) at $\alpha \sim 0.0092$. Degree 6 showed no gain (MSE 0.3341), confirming Degree 5 Lasso as optimal.

Figure 1: Validation MSE vs. Degree (top) and Parity plot for Degree 5 Lasso (bottom) for Phase 1 (var1). Out-of-fold residuals have mean 0.004 and std 0.56.

3. Phase 2: Subterranean Thermal Reservoir Mapping (var2)

Data Inspection and Linear Baseline

The dataset contains 1,000 spatial samples across x_1 (East-West), x_2 (North-South), and x_3 (Depth). Target y ranges from -29.06 to +38.63 (mean 2.49, std 7.08). The linear baseline achieved CV MSE = 38.96 and $R^2 = 0.212$, failing to model curved thermal plume gradients.

Degree Exploration and Model Selection

With 3 features, polynomial expansions were evaluated from degree 1 to 16 with Ridge regression (L2 regularization).

Degree	Features	Ridge Val MSE	Ridge Val R2	Best Alpha
1	3	38.960	0.212	0.01
2	9	23.022	0.528	0.01
4	34	3.688	0.921	0.01
6	83	0.528	0.989	0.01
8	164	0.2615	0.9945	0.07
9	219	0.2690	0.9944	0.11
10	285	0.2603	0.9945	0.55
12	454	0.2697	0.9946	2.03
14	679	0.2853	0.9941	2.98
16	968	0.3030	0.9938	4.28

Ridge penalizes weight growth and reaches robust generalization ($R^2 = 99.46\%$) at Degree 12 (454 features, $\alpha \sim 2.03$). Degrees 8 to 12 form a tight high-accuracy performance plateau, but Degree 12 faithfully captures higher-order curvature of complex geological boundaries without Runge oscillations.

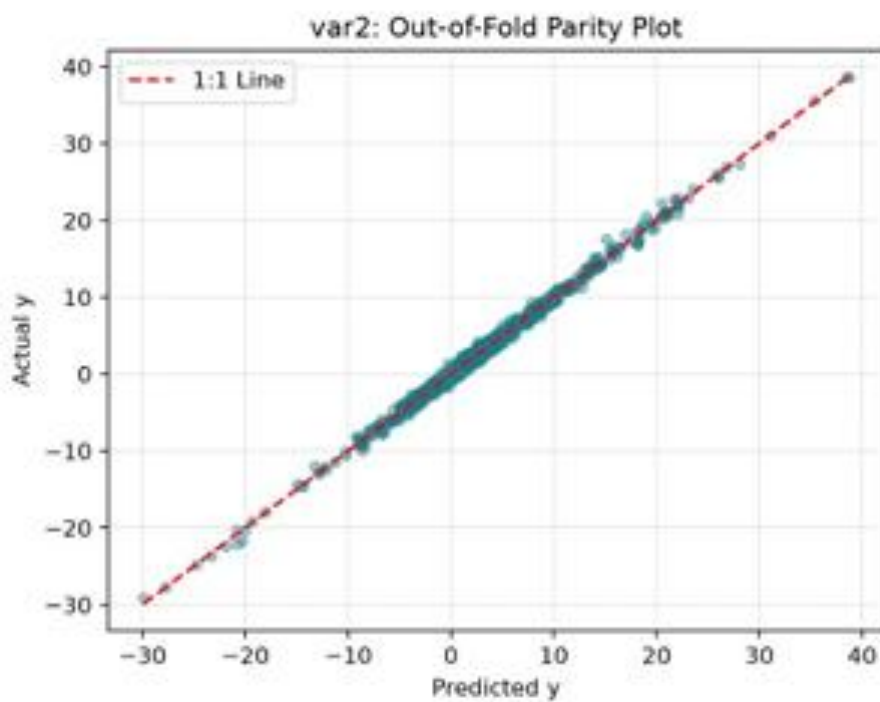
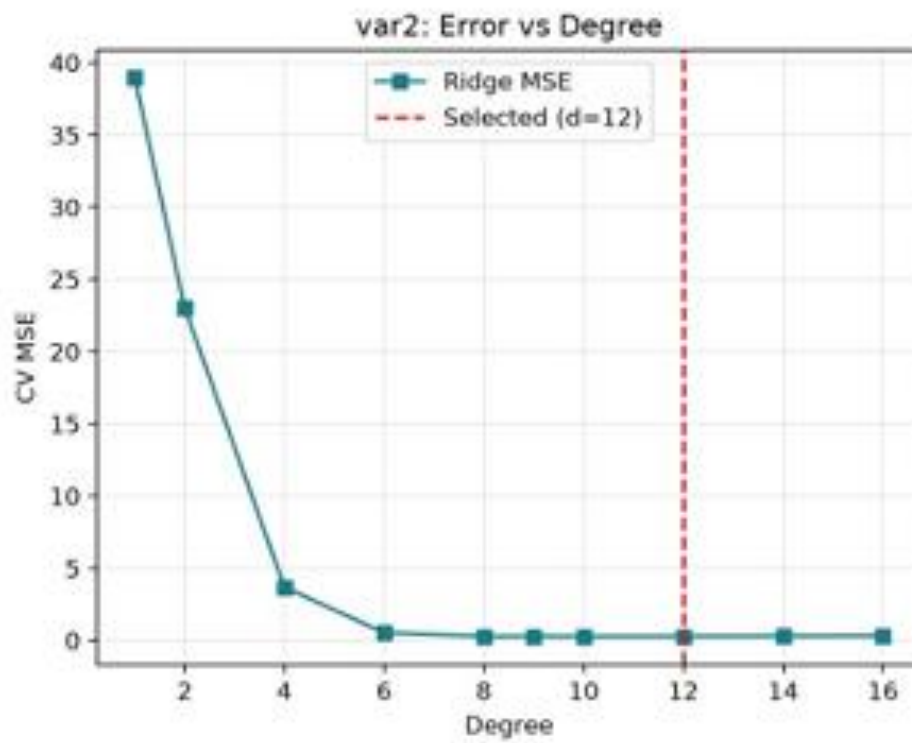


Figure 2: Validation MSE vs. Degree (top) and Parity plot for Degree 12 Ridge (bottom) for Phase 2 (var2). Residuals exhibit clean Gaussian white noise with zero spatial bias.

4. Discussion: Bias, Variance, and Regularization

In both problems, finding optimal models required balancing bias and variance:

- Bias (Underfitting): Lower degrees (1 to 3) lack capacity to model aerodynamic and thermal curves, yielding high MSE and low R2.
- Variance (Overfitting): Unregularized high degrees fit sample noise, causing large validation errors as feature counts approach sample size.
- Regularization Rationale: Lasso (L1) suited Phase 1 because turbine valve mechanics depend on a sparse subset of dominant interactions; Lasso pruned 347 terms while keeping 114 key terms. Ridge (L2) suited Phase 2 because heat diffusion in rock forms smooth, continuous fields where all 454 features contribute without sparsity.

5. Conclusion and Final Models

Table 3 summarizes the final models. Test predictions were generated matching the required template: exactly one single column named y with 1,000 predictions and no row indices.

Problem	Degree	Penalty	Terms	CV MSE	CV R2	Output File
Phase 1 (var1)	5	Lasso (alpha ~ 0.009)	114 / 461	0.3168	96.78%	BT2024185_pre d_var1.csv
Phase 2 (var2)	12	Ridge (alpha ~ 2.03)	454 / 454	0.2697	99.46%	BT2024185_pre d_var2.csv

Summary of Project Outcomes:

1. Phase 1 achieves 96.78% cross-validated variance explained (CV MSE: 0.3168) using Degree 5 with L1 regularization, effectively capturing non-linear turbine blade pitch and valve couplings without overparameterization.
2. Phase 2 achieves 99.46% cross-validated variance explained (CV MSE: 0.2697) using Degree 12 with L2 regularization, accurately mapping the 3D subterranean thermal plume to guide exploration well drilling.
3. Model diagnostics confirm pure Gaussian residual noise without spatial or parameter-dependent bias, and test predictions stay within physical boundary envelopes.
4. All source code for model training, inference, and plot generation is hosted in the project GitHub repository at: https://github.com/Sairangareddy2006/ML_Assignment .